# JUBORAJ AHMED

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## Education

**BRAC University** *Dhaka, Bangladesh*
*Bachelor of Science in Computer Science* 2026

**Overall GPA:** 2.98/4.00 **Final 60 Credits GPA:** 3.00/4.00

## Research Interests

Biomedical and Health Informatics • Trustworthy and Explainable AI in Medicine • Clinical Machine Learning • Electronic Health Record Data Analysis • Natural Language Processing for Healthcare • Multimodal and Patient-Facing AI Systems

## Research Experience

**Directional Constraints for Patient-Specific Alert Thresholds in Early Sepsis Monitoring: An Empirical Bayes Analysis**
*Completed Manuscript; preparing for journal submission* *2026*

- Investigated how patient-specific adaptation can alter clinical alert thresholds in early sepsis monitoring, with particular emphasis on **false-positive burden, sensitivity, and clinical utility** in decision-support settings.
- Studied a failure mode in which early disease-related risk may be misinterpreted as stable patient heterogeneity, potentially elevating alert thresholds and suppressing clinically useful early warnings.
- Developed and evaluated **Directionally Constrained Empirical Bayes (DC-EB)** threshold adaptation, restricting upward personalization while retaining patient-specific downward adjustment.
- Evaluated thresholding strategies on the **PhysioNet/CinC 2019 sepsis dataset** comprising **40,336 ICU patients**, using 8-, 12-, and 24-hour adaptation histories, patient-level data partitions, and **2,000 paired bootstrap replicates**.
- Found that directional constraints mitigated sensitivity and normalized clinical-utility losses associated with unrestricted personalization, although superiority over the optimized Global threshold was not conclusively established.

**Trustworthy Vision-Language Models for Medical Image Understanding**
*Ongoing Research Collaboration* *2026–Present*

- Investigating the reliability of medical vision-language models when image-grounded clinical evidence conflicts with misleading textual instructions, authority cues, or social pressure.
- Designing controlled experiments to examine **prompt-induced diagnostic capitulation**, assessing whether models abandon evidence-supported medical interpretations under conflicting contextual information.
- Studying how multimodal medical AI systems respond when visual evidence conflicts with textual context, with emphasis on **reliability, multimodal grounding, and trustworthy clinical decision support**.

**JASAL: A Remote Patient Health Monitoring System Driven by Federated Learning**
*Undergraduate Thesis, BRAC University; supervised by M. Mostakim* *2025–2026*

- Co-developed a federated learning framework for privacy-aware physiological anomaly detection using **32,796 records across 10 non-IID client partitions**, modeling decentralized healthcare environments.
- Designed adaptive patient profiling and reconstruction-error-based feature attribution to support personalized anomaly detection and interpretable physiological monitoring.
- Integrated differential privacy, secure aggregation mechanisms, and INT8 model quantization to investigate privacy-aware and resource-efficient healthcare AI.

- Evaluated the federated anomaly detection framework, with the global model achieving **96.28% accuracy** and **0.978 AUROC** in the reported experimental evaluation.

## Selected Research Projects

**PromptPoisonLab**

*Adversarial Prompt Injection and LLM Security Benchmarking* *2026*

*Python, FAISS, SentenceTransformers, SQLite, Docker*

- Developed an experimental framework for generating and analyzing adversarial prompts across **11 attack families**, supporting systematic investigation of instruction-following model vulnerabilities.
- Implemented evolutionary prompt mutation, embedding-based semantic deduplication, and structured experiment storage to support reproducible security evaluations and failure analysis.

**MediTrack**

*Machine Learning-Based Healthcare Risk Assessment* *2026*

*Python, Django, Scikit-learn, Pandas, NumPy*

- Developed a healthcare risk-assessment application using a Gradient Boosting classifier trained on the Pima Indians Diabetes dataset.
- Implemented a three-tier risk classification workflow and integrated model inference into a Django-based application.
- Evaluated predictive performance using class-specific F1 scores and macro-F1, achieving approximately **0.717 macro-F1** in the reported evaluation.

## Technical Skills

**Programming:** Python, C++, SQL

**Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NumPy, Pandas, SciPy

**Research Methods:** Clinical Machine Learning, Empirical Bayes, Statistical Bootstrapping, Federated Learning, Differential Privacy, Natural Language Processing, Transformers, Vision-Language Models, Model Evaluation, Explainable and Trustworthy AI

**Research Tools:** Git, GitHub, Docker, Linux, FAISS, Weights & Biases, Jupyter Notebook, Google Colab

**Scientific Writing:** LaTeX, BibTeX, IEEE/ACM manuscript preparation

## Additional Information

**TOEFL iBT (March 28, 2026):** Overall **5.0/6** (**CEFR C1**); ETS comparable score **100/120**.
**Section scores (1–6 scale):** Reading 4.5, Listening 5.0, Writing 5.5, Speaking 4.0.